

The uniform distribution

A continuous uniform distribution assigns equal probability to intervals of equal length within its range.

Are the intervals from 2 to 4 and from 6 to 8 equally likely under $U(0,10)$?

The range

Model: X is uniform from 2 to 14 minutes.

$$X \sim U(2, 14)$$

Which values are possible, and what is the width of the range?

Uniform density

The density is constant throughout the uniform range and zero outside it.

$$f(x) = \frac{1}{b - a} \quad (a \leq x \leq b)$$

What is the density for U(2,14)?

A waiting interval

Model: a shuttle wait is uniform from 0 to 12 minutes.

What is the probability of waiting between 3 and 7 minutes?

A short wait

Model: a shuttle wait is uniform from 0 to 12 minutes.

What is the probability of waiting at most 4 minutes?

A long wait

Model: a shuttle wait is uniform from 0 to 12 minutes.

What is the probability of waiting longer than 9 minutes?

A partially overlapping interval

Model: X is uniform from 2 to 14 minutes.

What is $P(0 < X < 5)$?

Outside the support

Model: X is uniform from 2 to 14 minutes.

What is $P(X > 15)$?

The center of a uniform distribution

The uniform mean is the midpoint of its range.

$$\mu = \frac{a + b}{2}$$

What is the mean for U(2,14)?

Uniform spread

For a uniform distribution, standard deviation depends on the width of the range.

$$\sigma = \frac{b - a}{\sqrt{12}}$$

**Which has greater standard deviation:
U(0,12) or U(20,32)?**

Calculating uniform spread

Model: a duration is uniform from 2 to 14 minutes.

What is its standard deviation, in minutes?

A percentile cutoff

Model: a shuttle wait is uniform from 0 to 12 minutes.

What wait time is the 75th percentile?

An upper-tail cutoff

Model: a shuttle wait is uniform from 0 to 12 minutes.

Only 10% of waits exceed what cutoff?

The middle half

Model: a shuttle wait is uniform from 0 to 12 minutes.

What interval contains the middle 50% of waits?

A conditional interval

Conditional probability restricts the possible outcomes. A wait is uniform from 0 to 12 minutes and has already exceeded 4 minutes.

What is the probability that the total wait exceeds 8 minutes?

The conditional density

Model: $X \sim U(0,12)$, in minutes. We know $X > 4$.

What new uniform density describes the restricted interval, and how does it give $P(X > 8 \mid X > 4)$?

Changing the range

Model A is $U(0,12)$. Model B is $U(0,24)$, with time measured in minutes.

How does doubling the range change the probability of a wait below 6 minutes?

Checking the model

Synthetic waits, in minutes: 0.4, 0.6, 0.7, 0.9, 1.1, 1.3, 9.8, 11.5.

What feature would you investigate before accepting a uniform model on 0 to 12?

Theoretical and empirical summaries

Synthetic sample: mean 8.2 min, SD 3.6 min.
Proposed model: $U(2,14)$, with theoretical mean 8 and SD about 3.46.

Must the sample summaries exactly equal the model parameters?

From one wait to many

Model: independent shuttle waits follow $U(0,12)$.

Would the average of 50 waits have the same distribution as one wait?